

MARK	SERVICE	SIZE	RATING	FACING	LOCATION
N1	Moist Bal Infeed	480x388	FLANGED	FLAT FACE	Top Hood Infeed End
N4A/B	Hot Air Supply	Ø x DN600	150#	RAISED FACE	Bottom Plenum Underside
N3A-C	Exhaust Air Outlets	3x DN500	150#	FLAT FACE	Top Hood Crown
N2	Dry Bal Discharge	DN350	150#	RAISED FACE	Discharge End Weir
N5	Plenum Fines Drain	DN100	150#	RAISED FACE	Plenum Invert Low Point
M1-M4	Inspection Doors	4x 500x400	HINGED	FLUID GASKET	Drive & Non-Drive Sides

MECHANICAL & PROCESS DESIGN SPECIFICATION – DR-101

1. Equipment Tag: DR-101 | Service: Continuous Vibrating Fluid Bed Bal Conditioning & Drying
2. Effective Deck Dimensions: 14,000 mm Length x 1,000 mm Width (Effective Area: 14.0 m2).
3. Material of Construction: All product contact parts SS304 (No. 4 Dull Lm sanitary polish). Non-contact IS2062 CS.
4. Fluidizing Air Rate: 14,150 kg/h (11,800 +5/+ 8 @ 85 deg C) | Superficial Velocity: 1.15 m/s (Unf = 1.85 m/s).
5. Water Evaporation Duty: 4012.1 kg/h (402.1 Thermal Heating Duty: 3124 kWth via Steam Coil HC-101.
6. Vibration Dynamics: Stroke amplitude: 3.5 mm @ 16 Hz (960 RPM). Twin unbalance motors 2 x 3.1 kW @ 45 deg.
7. Isolation Efficiency: 8x Heavy-duty helical coil springs with rubber snubbers (>96% dynamic isolation).
8. Residence Time: 18.3 – 22.3 minutes plug-flow with adjustable discharge overflow retention weir.
9. CIP / Washdown: Complete internal spray ball system with sanitary steam valves for wet cleaning cycles.
10. Safety Hold Final: dynamic balancing, spring resonance validation and FAT testing required prior to site release.

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.	
15,000 kg/day Solid Pigeon Pea Optimization Greenfield Project	
EQUIPMENT GENERAL ARRANGEMENT – FLUID BED DRYER DR-101	
PROCESS & MECHANICAL	STAGE: CONCEPTUAL FEED
DWG NO: DWG-FIN-IST-008	REV: V4.0
SHEET: 8 OF 12	
SCALE: AS SHOWN	DATE: 2026-03-05
STATUS: PRELIMINARY	DATE: 0-03-2026

CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT